Deep Learning-Based Kidney Disease Classification from MRI Images Using CNNs

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KIDNEY PATIENT CARE
Using CNNs · Class 2026

01 INTRODUCTION

- Kidney diseases affect 10–15% of the global population; CKD alone claims millions annually.
- MRI offers superior soft-tissue contrast for renal assessment — no ionizing radiation required.
- Manual MRI interpretation is labor-intensive and subject to inter-observer variability.
- AI-based CNNs can provide rapid, objective diagnostic support in resource-limited settings.

02 PROJECT OBJECTIVES

- Curate & preprocess a labeled kidney MRI dataset: Normal, Cyst, Tumor, Stone.
- Design and train a CNN via transfer learning (ResNet50 / EfficientNetB3 / DenseNet121).
- Build a web application for clinicians to upload MRI images and get predictions.
- Evaluate with Accuracy, Precision, Recall, F1-Score, and AUC-ROC metrics.
- Integrate Grad-CAM heatmaps for explainable AI-assisted diagnosis.

03 RESEARCH QUESTIONS

- Can CNNs accurately classify Normal, Cyst, Tumor, and Stone from kidney MRI?
- Which architecture — ResNet50, EfficientNetB3, or DenseNet121 — performs best?
- Does Grad-CAM explainability meaningfully improve clinical trust in AI predictions?
- How does transfer learning affect performance on limited MRI training data?

04 SCOPE & LIMITATIONS

Scope:

- Four-class classification (Normal, Cyst, Tumor, Stone) — clinical decision-support only.
- Full ML lifecycle: data acquisition → CNN training → web application prototype.

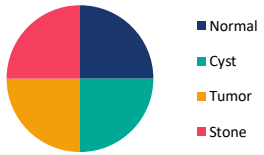
Limitations:

- Performance bounded by dataset size & diversity; may not generalize across MRI manufacturers or rare renal pathologies beyond the four defined classes.

05 METHODOLOGY

Dataset & Preprocessing

- Source: TCIA + open-source renal imaging repositories.
- 4 Classes: Normal · Cyst · Tumor · Stone
- Split: Train 70% | Val 15% | Test 15%
- DICOM→PNG, normalize [0,1], resize 224×224 px, augmentation.



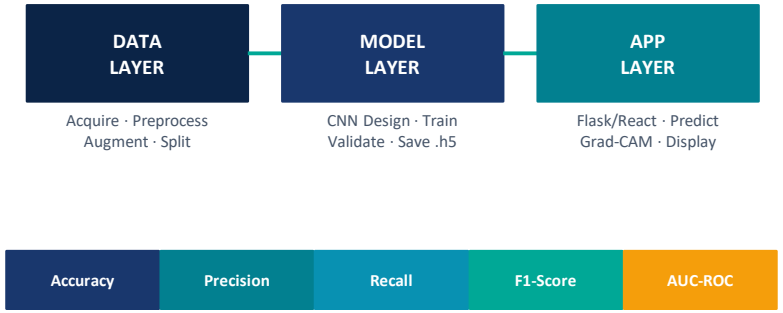
CNN Model (Transfer Learning)

- Architectures: ResNet50 · EfficientNetB3 · DenseNet121 (ImageNet pre-trained).
- Phase 1 – Feature Extraction: Freeze base; train custom head (GAP → Dropout 0.5 → Softmax ×4).
- Phase 2 – Fine-Tuning: Unfreeze top layers; retrain with low learning-rate on MRI data.
- Optimizer: Adam | Loss: Categorical Cross-Entropy | EarlyStopping (patience=10).

Tools & Libraries

- ML Stack: TensorFlow/Keras · OpenCV · NumPy · Scikit-learn · Matplotlib
- Backend: ASP.NET Core 8 · FastAPI (AI Service) | Frontend: React 18 + TypeScript

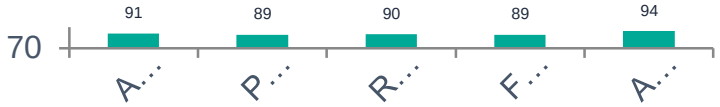
06 SYSTEM ARCHITECTURE



07 KEY SYSTEM FEATURES

- Multi-class CNN: Single model classifies Normal, Cyst, Tumor, and Stone from kidney MRI.
- Transfer Learning: EfficientNetB3 fine-tuned on MRI data — expected accuracy ~90%+.
- Grad-CAM Explainability: Heatmap overlay highlights regions most influential in the prediction.
- Role-based Access: Patient · Doctor · Admin with JWT authentication.
- Real-time Features: SignalR-powered chat, AI analysis notifications, appointment management.

08 EXPECTED RESULTS



09 RECOMMENDATIONS

- Expand dataset with multi-institution MRI data to reduce bias and improve generalization.
- Conduct prospective clinical validation alongside expert radiologists.
- Extend classification to rare/complex renal pathologies beyond four defined classes.
- Pursue regulatory approval (FDA 510(k) / CE marking) for clinical deployment.
- Integrate with hospital PACS and EHR systems for seamless clinical workflow embedding.

10 REFERENCES (IEEE Format)

[1] Z. Khene et al., "CNN-based renal tumor classification from contrast-enhanced CT," Eur. Urol., 2020.

[2] D. Reddy et al., "Staging CKD via ultrasound texture features using CNN," CBMS Conf., 2019.

[3] Z. Lin et al., "3D CNN on multiparametric MRI for ccRCC subtype classification," Radiology, 2021.

[4] H. Coy et al., "2D CNN on T2-weighted MRI for Bosniak renal mass classification," JACR, 2022.

[5] Y. Huo et al., "U-Net-based kidney & tumor segmentation from MRI," Med. Image Anal., 2019.